Most resilient receiver: withstanding GNSS jamming and spoofing

JammerTest in Bleik, Andoya, Norway ©David Jensen

Results from JammerTest in Norway - Septentrio regilarily validates and improves r of its GNSS core technology

Across the globe there is a growing need for improved resilience of PNT (Positioning, Navigation, Time) systems. president signed an Executive Order (https://www.dhs.gov/news/2020/02/12/dhs-statement-president-s-executi strengthen-national-resilience-through) to strengthen national resilience through PNT services. Geo-political ten demand higher security for operations in areas of navigational warfare (NAVWAR). For many GNSS applications

reference networks, drone surveillance or delivery as well as timing synchronization the consequences of PNT d
or loss can be dire. That is why GNSS receivers used in these mission-critical applications need to have the highe
resilience for robust operation in challenging environments.

AIM+ anti-jam and anti-spoof works (/node/912) against live GNSS interference

Septentrio provides receivers with the highest level of resilience against intentional and unintentional GNSS jam
(https://www.septentrio.com/en/learn-more/advanced-positioning-technology/aim-jamming-protection) and spo
(https://www.septentrio.com/en/learn-more/advanced-positioning-technology/aim-spoofing-protection)*.Testin
receiver technology under live interference conditions is critical to validate and continuously improve anti-jam a
spoof algorithms. That is why Septentrio participates in JammerTest, an event organized by the Norwegian gove
the remote island of Andøya, where live interference testing takes place in a controlled environment. Although S
regularly participates in such tests, many of these events are classified, and results cannot be shared publicly. Ja
is one of a few public events of this type where sharing or results is encouraged.

After 5 days of intensive testing in Norway, Septentrio's proprietary AIM+ anti-jamming and anti-spoofing techno
(/en/learn-more/advanced-positioning-technology/aim-resilient-and-secure-gnss/gps-receivers) proved to work
interference conditions. Test results revealed that under real interference receiver technology plays a key role, w
antenna technology plays a supporting role. By testing the receiver under various types of spoofing attacks it wa
that the best spoofing protection lies in having multiple anti-spoofing mechanisms working together.

Saab, a major aerospace and defense company, also took part in the JammerTest event and validated resilience
Septentrio receivers which are used in its demanding air traffic management system.

> Although they may not realize it, each year hundreds of millions of passengers rely upon Saab's air traffic management systems to get where they are going safely. Most of these systems in turn rely upon GPS/GNSS for positioning and time synchronization. With so much at stake, Saab is proud to offer industry leading anti-spoofing and anti-jamming capabilities using Septentrio equipment. Saab and Septentrio's recent collaboration at JammerTest showcased just how effective we can be together.
>
> **ROBERT BROWN**
> **DIRECTOR OF PORTFOLIO, SALES AND STRATEGY, SAAB AIR TRAFFIC MANAGEMENT**

Results from JammerTest 2024

During the 5-day event in

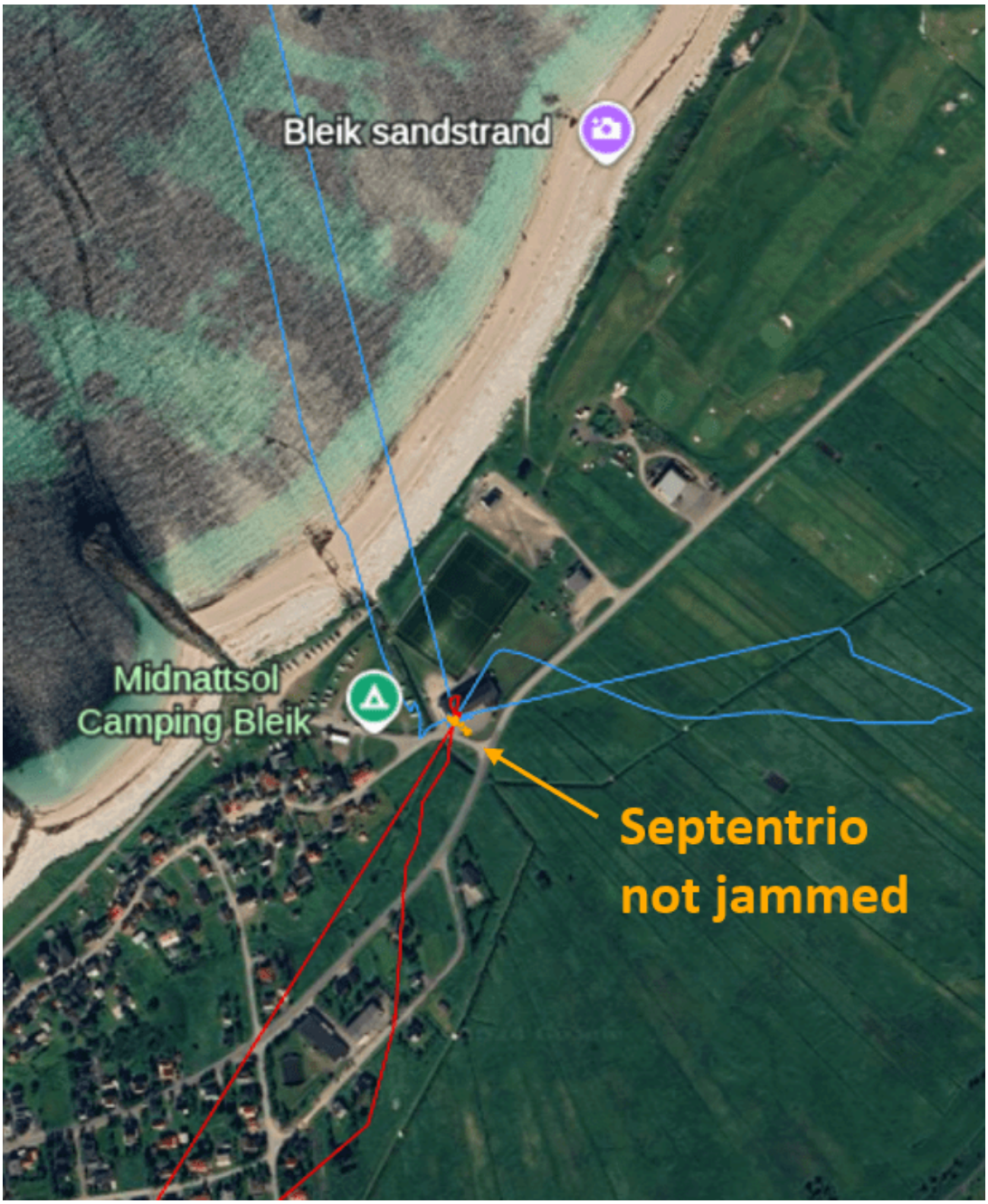

Figure 1: Septentrio receiver (orange) is not jammed even by very powerful interference, while two competitor receivers (red, blue) show

Norway Septentrio receivers were tested under various interference scenarios alongside several competition re
This image illustrates a static test in which multiple receivers were evaluated under extremely powerful jammin
conditions — signals that were 10 million times stronger than GNSS signals. The blue and red lines represent tw
competitor receivers, both of which delivered positions that were offset by hundreds of meters due to the impa
jamming. In contrast, the Septentrio receiver, shown in orange, demonstrated exceptional resilience to GPS / G
jamming. It consistently delivered accurate GNSS positioning while actively detecting and signaling the presenc
jamming.

In real-world scenarios, losing GNSS positioning can severely disrupt operations, leading to costly delays, opera
inefficiencies, and reputational damage. For instance, drones affected by jamming may lose control and crash.
advanced drones equipped with autopilots may enter fallback modes such as “hover,” “land,” or “circle,” or swit

alternate sensors like Inertial Navigation Systems (INS). However, without reliable absolute positioning, these se
prone to drift over time.

Even in GNSS spoofing attacks, jamming is often initially used to break the receiver's "lock" on GNSS signals to ir
chances of the spoofed signals being accepted during reacquisition.

Spoof-proof GNSS receivers

The image below shows a meaconing test where Septentrio receiver performance is compared to a high-end co
receiver. Meaconing is a form of spoofing where GNSS signals are recorded and then retransmitted with a time
spoofed receiver then "believes" it is at the location of the antenna that was used to record the GNSS signal. Dur
as the car traveled along a coastal road, the Septentrio receiver (orange line) maintained accurate positioning. Ir
the competitor receiver (red line) was spoofed, reporting its location on a mountain - the site of the interference
transmission antenna used in the JammerTest, as shown in this article's cover image.

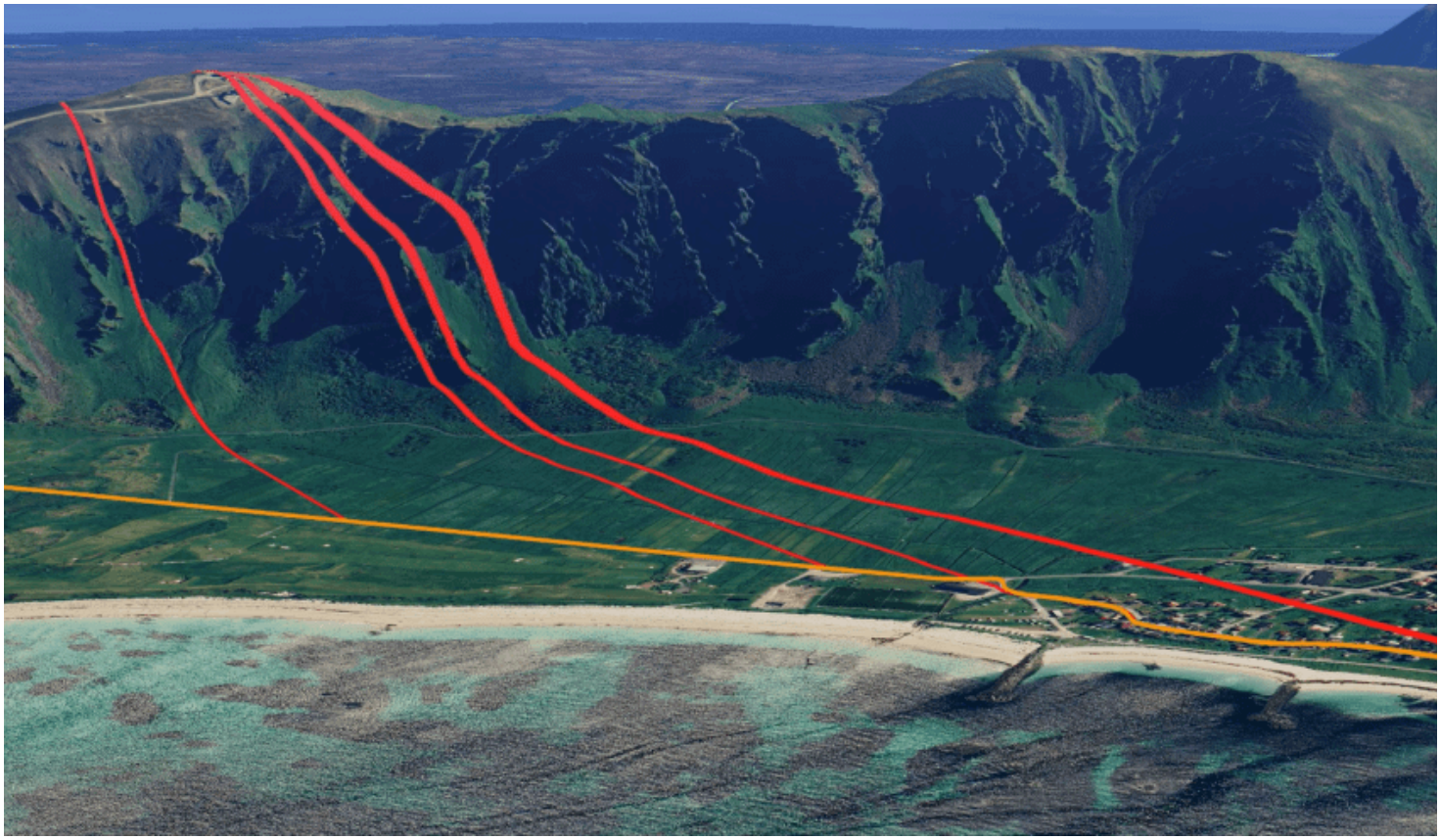

Figure 2: Car test shows that a Septentrio receiver (orange) is resilient to GNSS spoofing of type meaconing, while competitor receiver is spoofed to th

Danger: Spoofed positions without a warning

This graph shows a summary of performance of various receivers during spoofing testing throughout the 5-day
green sections represent the percentage of correct positions or successful spoofing mitigation. Grey indicates t
percentage of time the receiver reported no position during spoofing, and red shows the percentage of time th

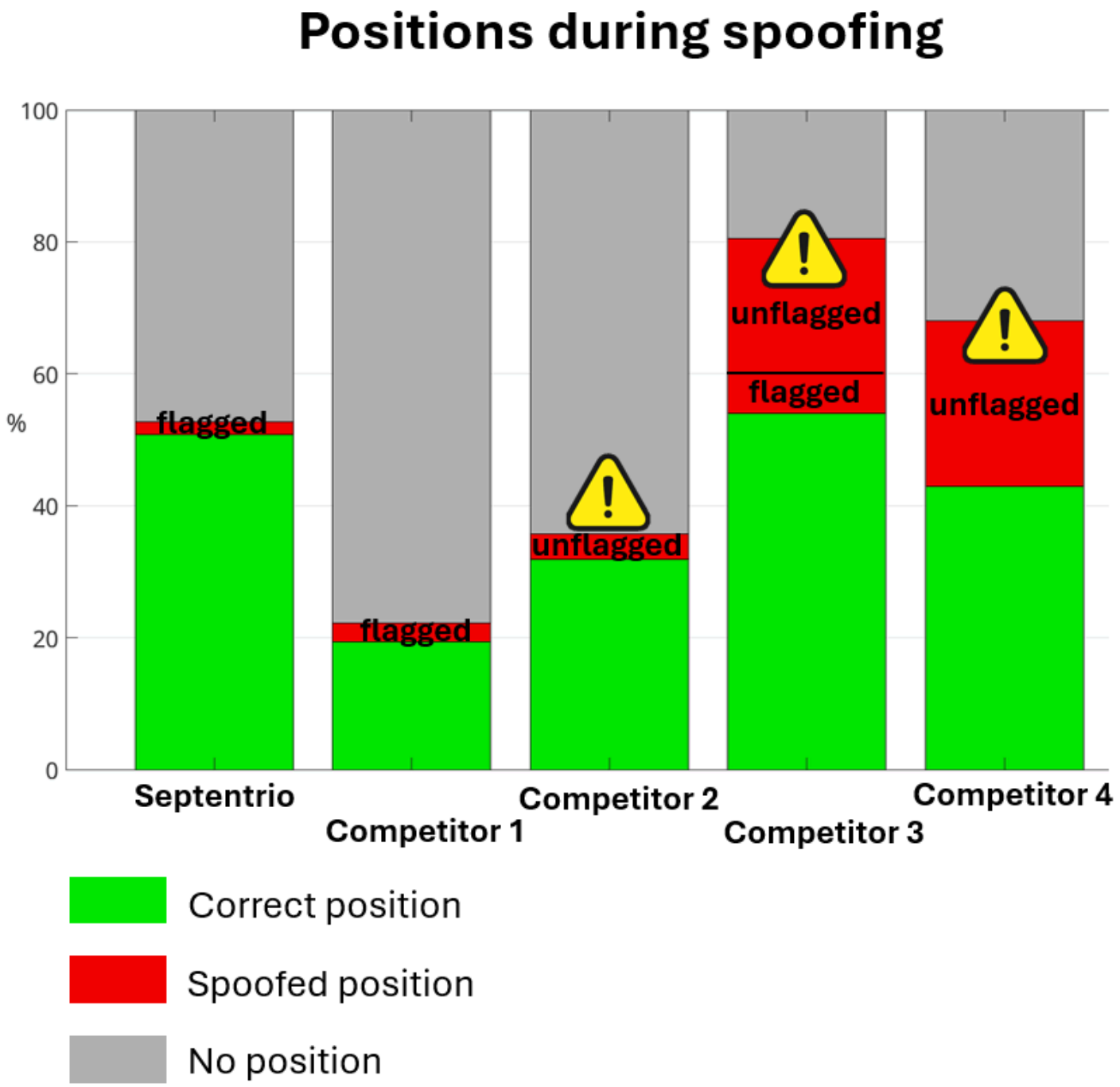

Fig. 3 : The combination of high GNSS spoofing mitigation (green) & spoofing detection (red, flagged) demonstrates that Septentrio receivers are relia various types of spoofing attacks, as compared to 4 other RTK competitor receivers.

displayed spoofed positions.

The Septentrio receiver showed minimal spoofed positioning, even under sophisticated spoofing attacks, and c detected and flagged spoofing when it occurred, demonstrating situational awareness. In contrast, three comp receivers failed to flag spoofing in many cases, displaying erroneous positions without warning the system of p risks.

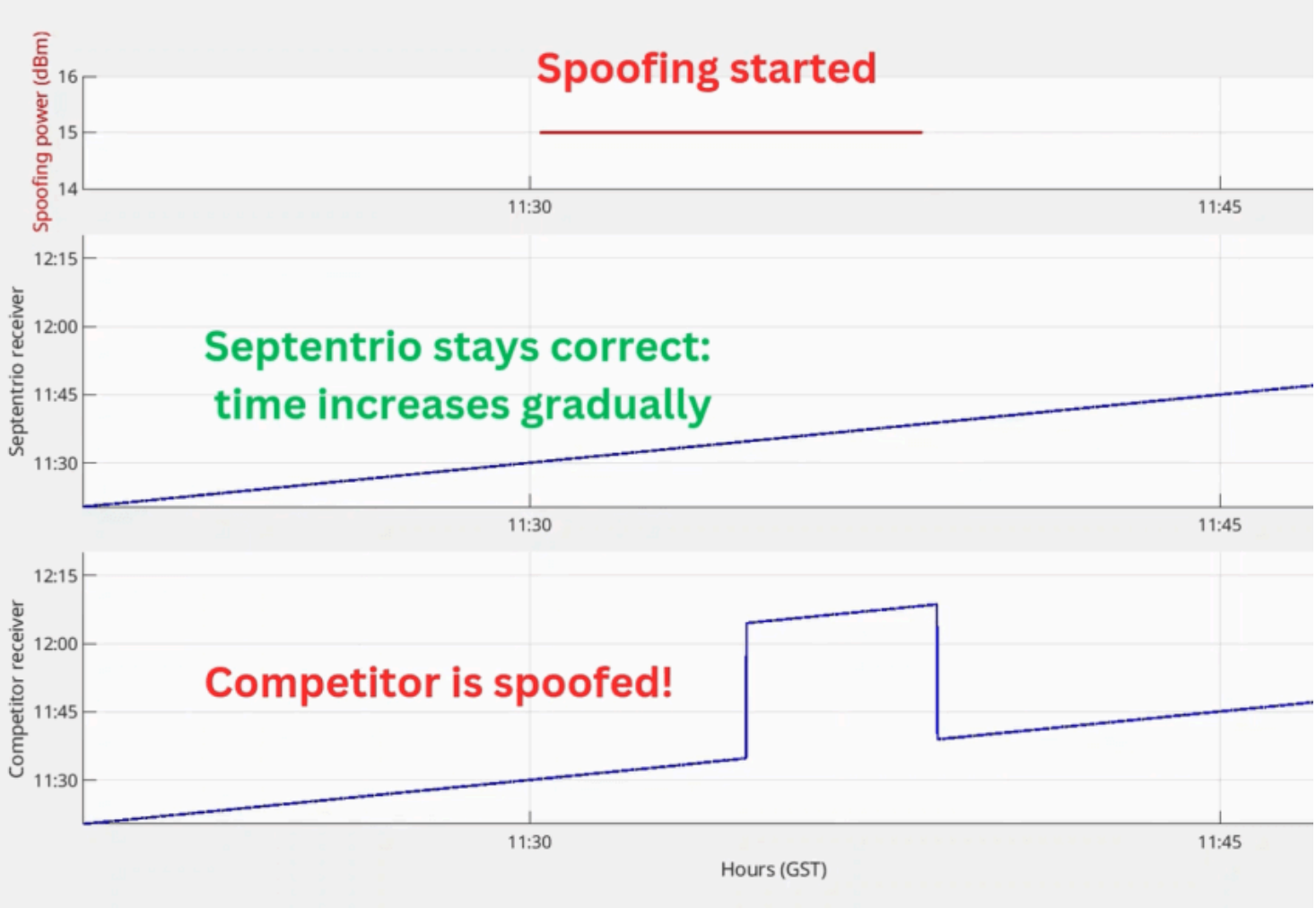

Fig. 5 : As GNSS time is spoofed, the Septentrio receiver keeps delivering correct timing information, while a competition receiver shows incorrect tim

Resilience of GNSS timing validated

The plot below shows results from a test where GNSS time was spoofed. The Septentrio receiver maintained ac timing, while a competitor receiver was spoofed and displayed incorrect time. Applications such as telecom, po and financial institutions depend on precise GNSS timing. Malicious spoofing can result in outages, with total cc reaching millions of dollars or euros. Investing in resilient GNSS technology is essential to protect mission-critic and the users who depend on them.

Multi-layered receiver protection pays off

During JammerTest 2024, Septentrio receivers with AIM+ technology demonstrated exceptional resilience to hig jammers and various types of spoofing. This performance reflects over 25 years of expertise in developing reliab positioning technology, incorporating multiple layers of protection:

- Multi-frequency (https://www.septentrio.com/en/learn-more/about-GNSS/why-multi-frequency-and-multi-con matters) technology for signal diversity, ensuring fallback during jamming or spoofing

- Cryptographic signal authentication (OSNMA (https://www.septentrio.com/en/learn-more/insights/osnma-late anti-spoofing-security)) to verify genuine signals
- Anomaly detection powered by continuously updated big data from years of field observations

As seen on the news (https://www.forbes.com/sites/marisagarcia/2024/02/03/faa-tells-pilots-to-go-analogue-as-spoofing-incidents-increase/), GNSS spoofing is increasingly common due to readily available transmitter hardw signal simulator software. Spoofing poses an even greater risk than jamming, as it can not only disrupt operatio manipulate navigation systems, potentially resulting in planned collisions or theft (https://www.csmonitor.com/World/Middle-East/2011/1215/Exclusive-Iran-hijacked-US-drone-says-Iranian-engineer). Securing your system at the receiver core is the most effective strategy to ensure resilient, reliable op

Learn more about AIM+ Anti-Spoofing (https://www.septentrio.com/en/learn-more/advanced-positioning-techn anti-spoofing-protection) technology or download our comprehensive GNSS Spoofing brochure (https://www.septentrio.com/en/learn-more/brochures/gps/gnss-spoofing-brochure). The video below is a sumr Septentrio's results from JammerTest 2024. Feel free to share this video with those interested in resilient and re positioning.

Tried, tested and true

Always accurate and available Positioning, Navigation and Time or Assured PNT (/en/applications/assured-pnt-space/assuredpnt) is key to successful industrial or critical operations in challenging environments. By regularly participating in live events such as the JammerTest, Septentrio's anti-jamming and anti-spoofing technology is c being tested and improved to withstand the latest interference attacks. This technology has also been confirme effective by users out in the field, who are using Septentrio receivers in places of malicious interference, such as contested borders.

Although an anti-jam antenna can play a role in hardening the system, the most effective approach to securing is to use a receiver with advanced anti-spoofing and anti-jamming (AJAS) protection. For spoofing this means inc several layers of protection which shield the receiver from various types of attacks. Septentrio receivers (/en/pr receivers) come in various formats from rugged boxes to OEM boards and compact modules, like the low-weigh power mosaic-X5 module receiver, which is also easy to integrate and is compatible with popular autopilots like ArduPilot and PX4 Autopilot. For more information about resilient GNSS receivers and technology please contac (/en/contact).

AIM+ detects and mitigates GNSS jamming

The figure on the right shows that during jamming a Septentrio GNSS receiver in a car delivered accurate positi along the whole route (orange track), while a high-end GNSS/INS competition receiver lost positioning two time track).

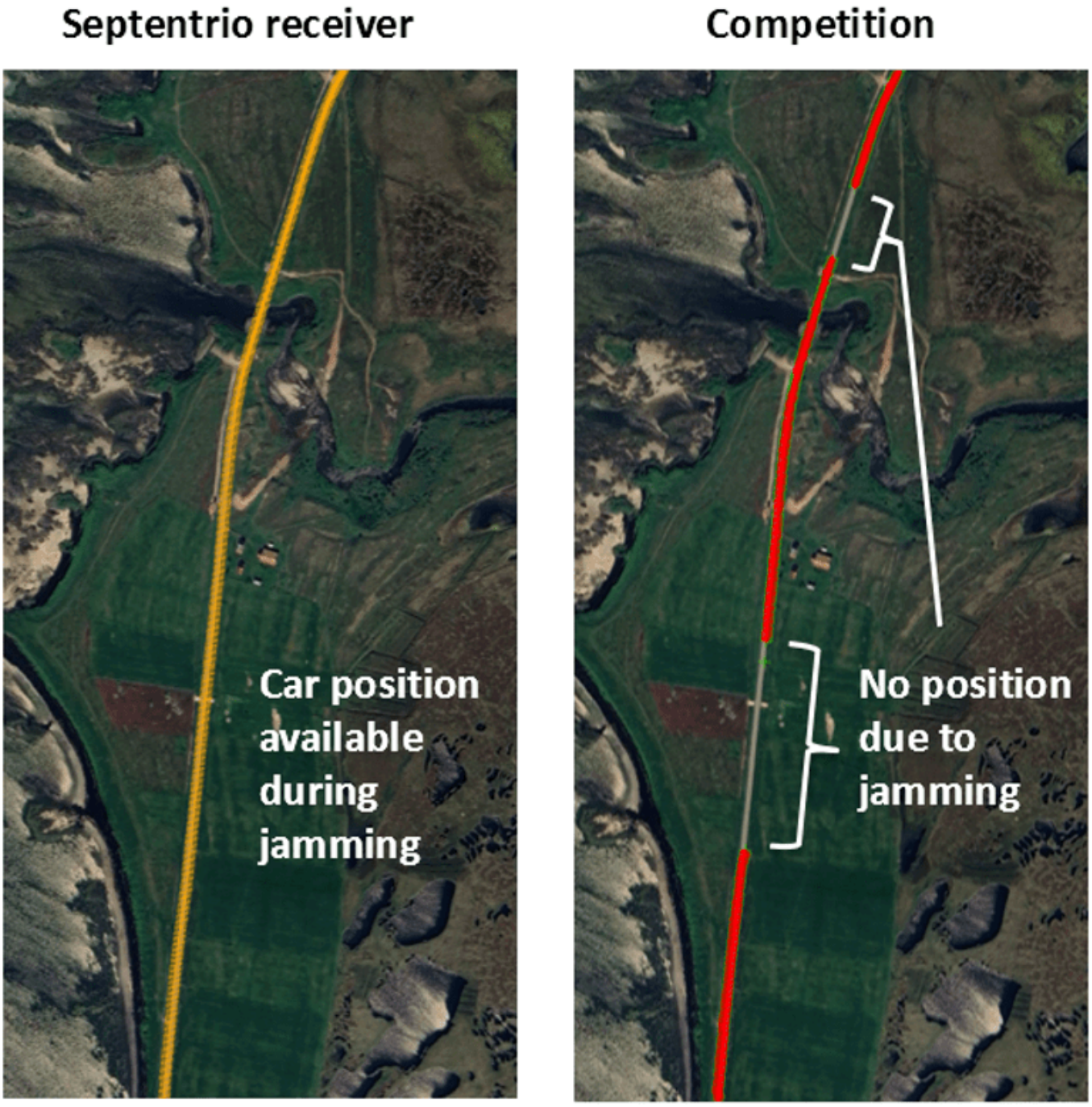

Figure 1 In real-time GNSS interference, the Septentrio receiver on the test car can provide accurate positioning, while the competitor receiver will los after a period of time.

Figure 2: During the jamming mobile car test a common "cigarette lighter"-type jammer was used. Image credit: Testnor.

During this test a “cigarette” jammer was used, which is one of the most common types of jammers and can ea purchased online. It emits signals with power between 10-15 dBm and jams GPS L1, L2 signals. Other jamming included highly powerful jammers with signals 10 million times more powerful than GNSS signals.

The plot below shows results of a mobile jamming test performed with the “cigarette” low-power jammer. The s noise time plot shows that when jamming is started the quality of the tracked GNSS signals decreases. A Septer receiver, shown at the top continues to track the signals and to output positioning, while a competition receive positioning for a period of time.

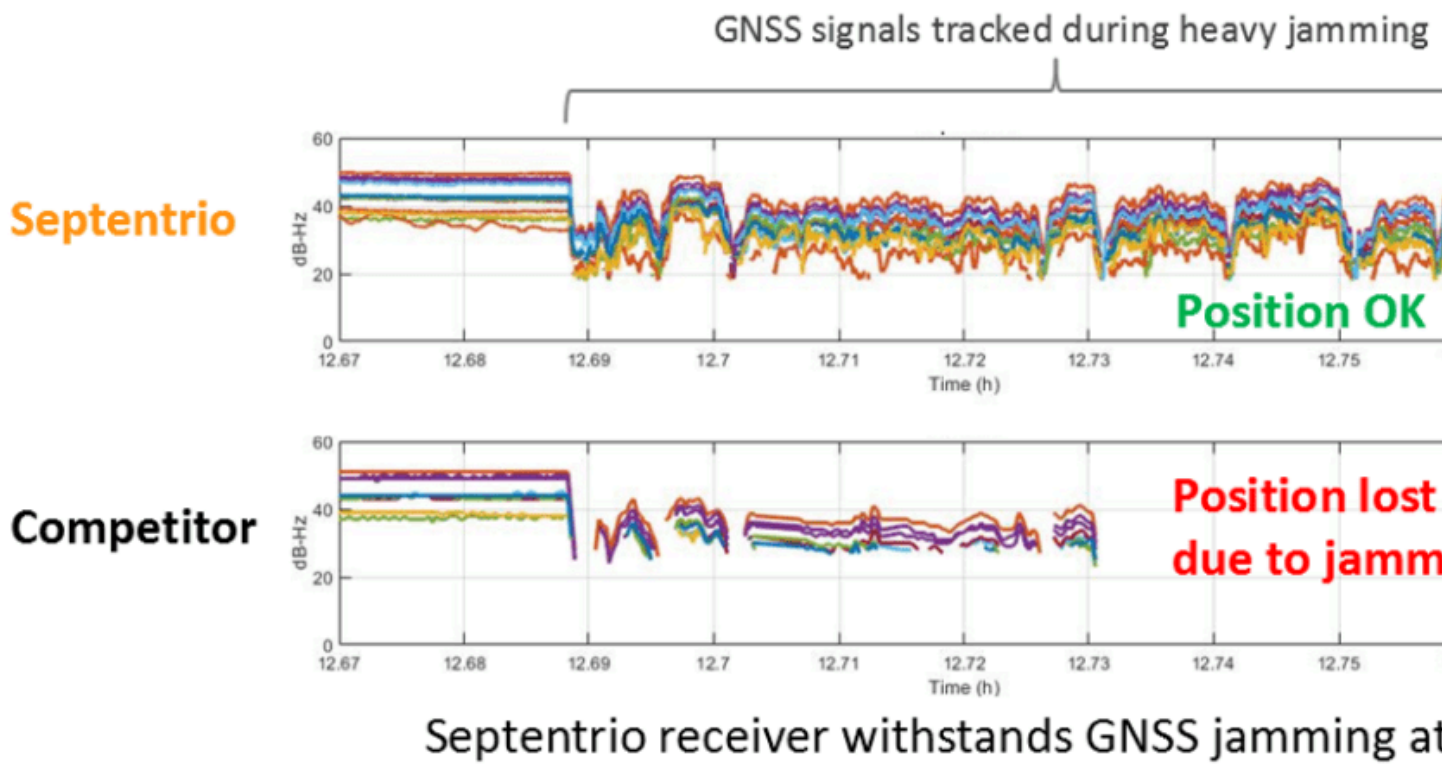

Figure 3: As GNSS signals start getting jammed, their quality decreases as seen in this signal-to-noise plot. The Septentrio receiver maintains tracking while the competition receiver loses positioning for a period of time.

Over the 1-day period of intensive jamming tests, receivers with AIM+ demonstrated 99,5% positioning availabili various forms of jamming from simple continuous narrow-band interference to the most complex wide-band transmissions.

The magic is in the GNSS receiver

For mission-critical applications an anti-jam antenna can play a role in achieving maximum resilience against RF interference. During the JammerTest, 3 receivers were tested under heavy multi-frequency wideband jamming i combination with antennas of varying sophistication. The signal-to-noise time plots below show signal quality du test. In the top plot, a receiver with a standard wideband helical antenna does not have AIM+ anti-jamming tech so immediately loses tracking of satellite signals during jamming. A receiver with the same antenna, but with AIM continues to track signals and deliver positioning (middle plot). The bottom plot shows a receiver with AIM+ cou anti-jam antenna. The drop in signal quality is slightly less than with a standard antenna and the receiver contin signals and to deliver positioning.

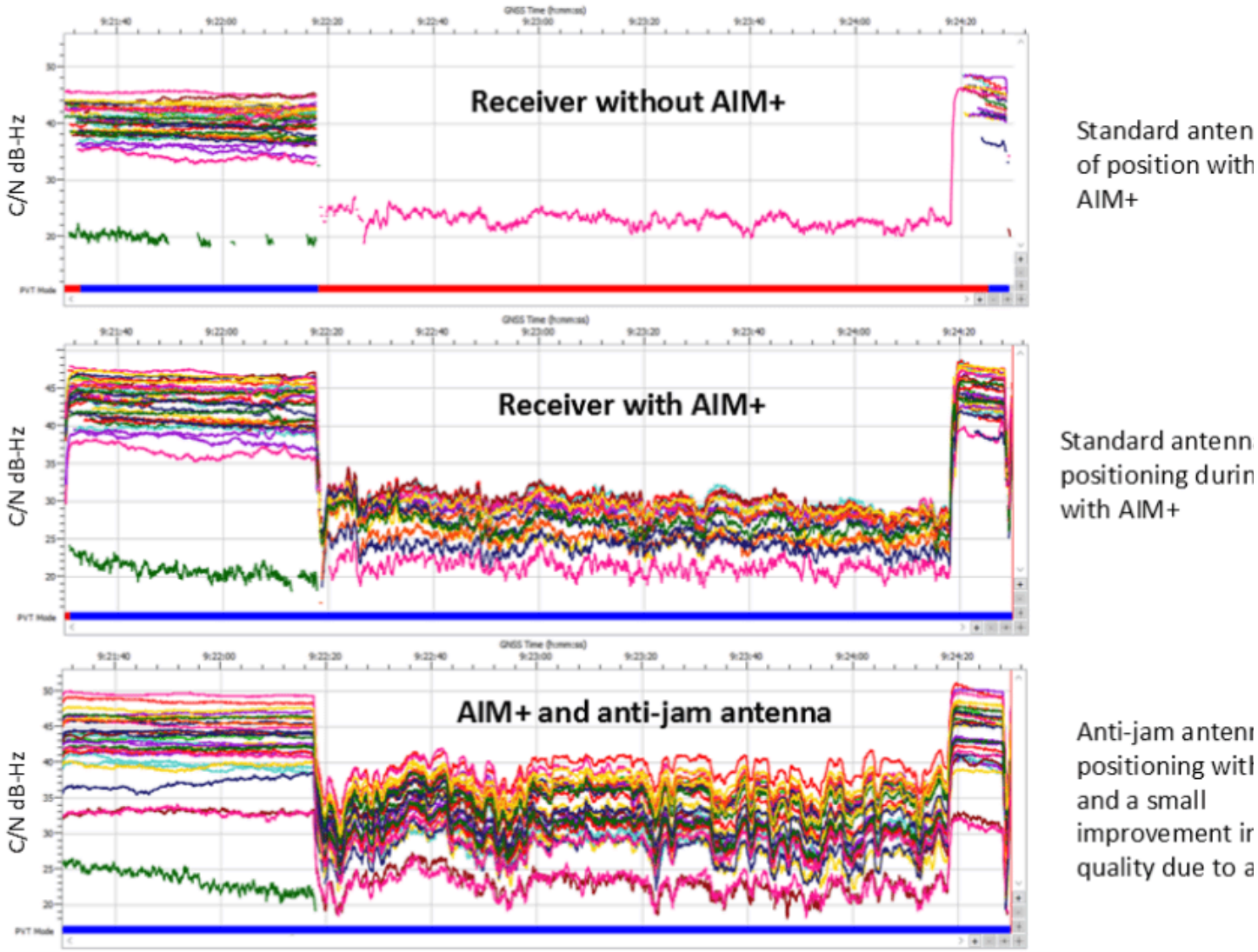

Figure 4: During heavy jamming Septentrio receiver with AIM+ mitigates interference and provides positioning (middle plot). With anti-jam antenna quality improvement and positioning (bottom plot). Without AIM+: no positioning (top plot)

Tests with various anti-jam antennas showed interference reduction of about 10 dB due to the antenna. While a anti-jam technology like AIM+ plays a key role in positioning availability under jamming, an antenna plays a supp and can improve the chances of getting positioning in cases where the jamming is still slightly stronger than the the receiver to mitigate it. While anti-jam antennas can be effective for wide-band “white-noise” jamming they ar effective for other types of jamming.

Comprehensive anti-spoofing requires multiple layers of protection

Hackers employ various types of spoofing to hijack positioning, navigation or time, from simple asynchronous sp sophisticated synchronous types which are almost identical to real GNSS signals. For the highest degree of anti-s protection and situational awareness a receiver requires several layers of protection (find out more in our GPS S brochure (/en/learn-more/brochures/gps/gnss-spoofing-brochure)).

AIM+ detects, flags and mitigates GNSS spoofing

The plots below show positioning of several different receivers in a moving car during a GNSS spoofing attack.

During a live spoofing attack at the JammerTest 23 in Norway a Septentrio receiver in car stayed on track along the coastal road (orange track in eac competitor receivers (blue, green, red track) were spoofed and show erroneous positions.

In the plots shown in the video, a Septentrio receiver (orange tracks in each image) mitigates GNSS spoofing anc accurate positioning along the coastal road, while a competition receiver (blue track on the left) is spoofed and i "flight" straight over the mountains. The plot in the middle shows another competition receiver (green) which in same spoofed trajectory as the blue track, but with outliers. The right plot shows a third competition receiver (re strongly affected by spoofing and delivers erroneous positions, which are different from those transmitted by th During this spoofing attack the Septentrio receiver was able to stay on track because of its in-depth multi-freque capability, which allowed it to mitigate spoofing by switching to other "healthy" signals of alternative constellatio

Septentrio receivers consistently detect GPS spoofing

During various live spoofing tests in Norway a large amount of data was collected from various receivers over se The graph below shows that unlike competition, Septentrio receivers consistently detected and flagged spoofing and flagged all cases where spoofing would have resulted in large positioning errors. Spoofing which "places" th away from the real position is especially dangerous since it can result in a collision, system damage or comprom bystander safety.

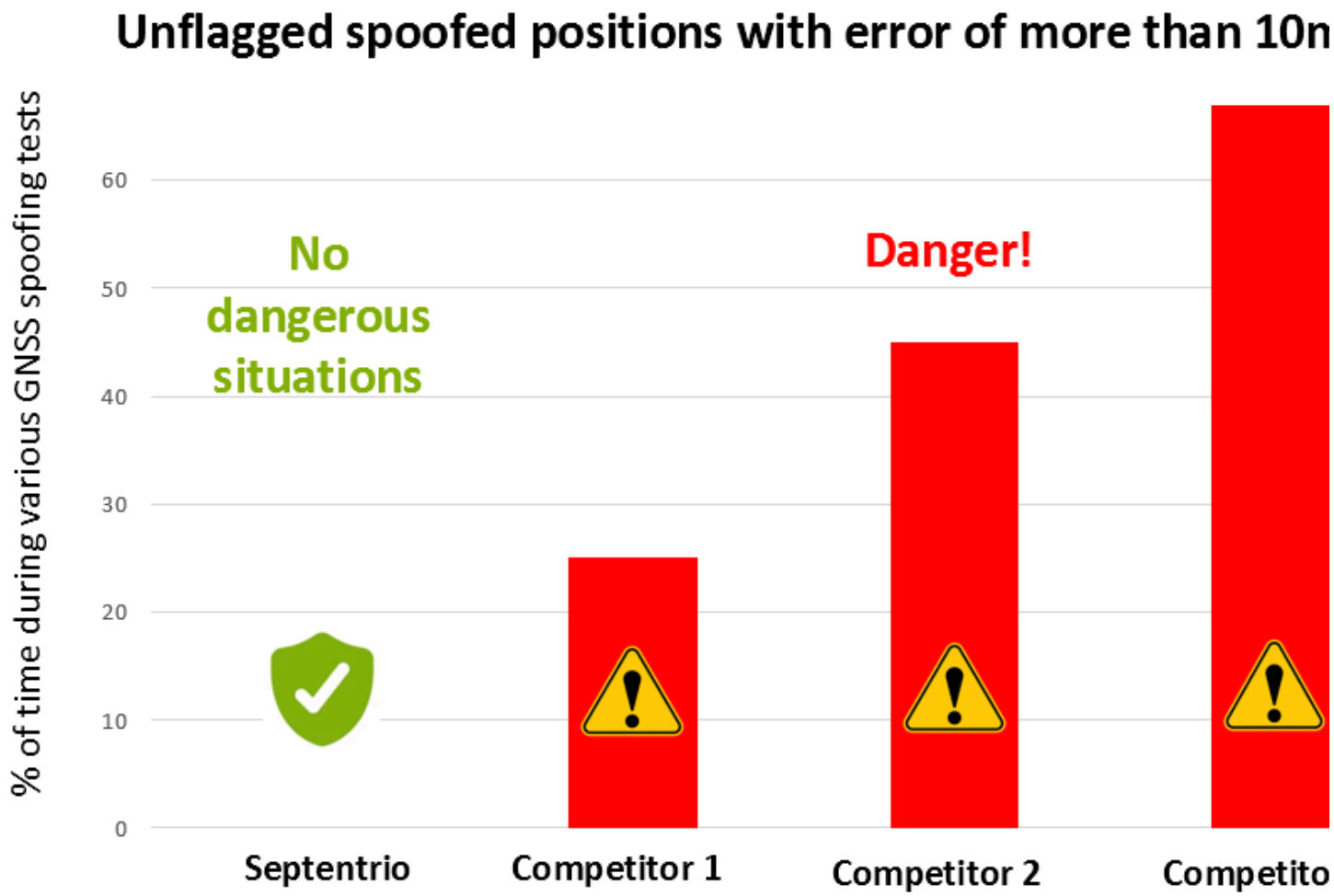

Figure 6 The red bars indicate undetected spoofed positioning with large errors. Septentrio receivers use several mechanisms simultaneously to dete flag all spoofed positions with large errors.

The spoofing flag of competition receivers did not go on during any of the spoofing attacks seen at the JammerT The graph shows that for a significant percentage of time these receivers reported positions which were off by r 10m, due to spoofing. During other periods the errors were smaller than 10m or no position was reported at all

OSNMA Galileo signal authentication detects GNSS spoofing

Another layer of anti-spoofing protection is Navigation Message Authentication (NMA), which leverages cryptogr techniques to detect falsely generated signals. The image below shows Galileo's OS-NMA (https://www.septentrio.com/en/learn-more/insights/osnma-latest-gnss-anti-spoofing-security)at work.

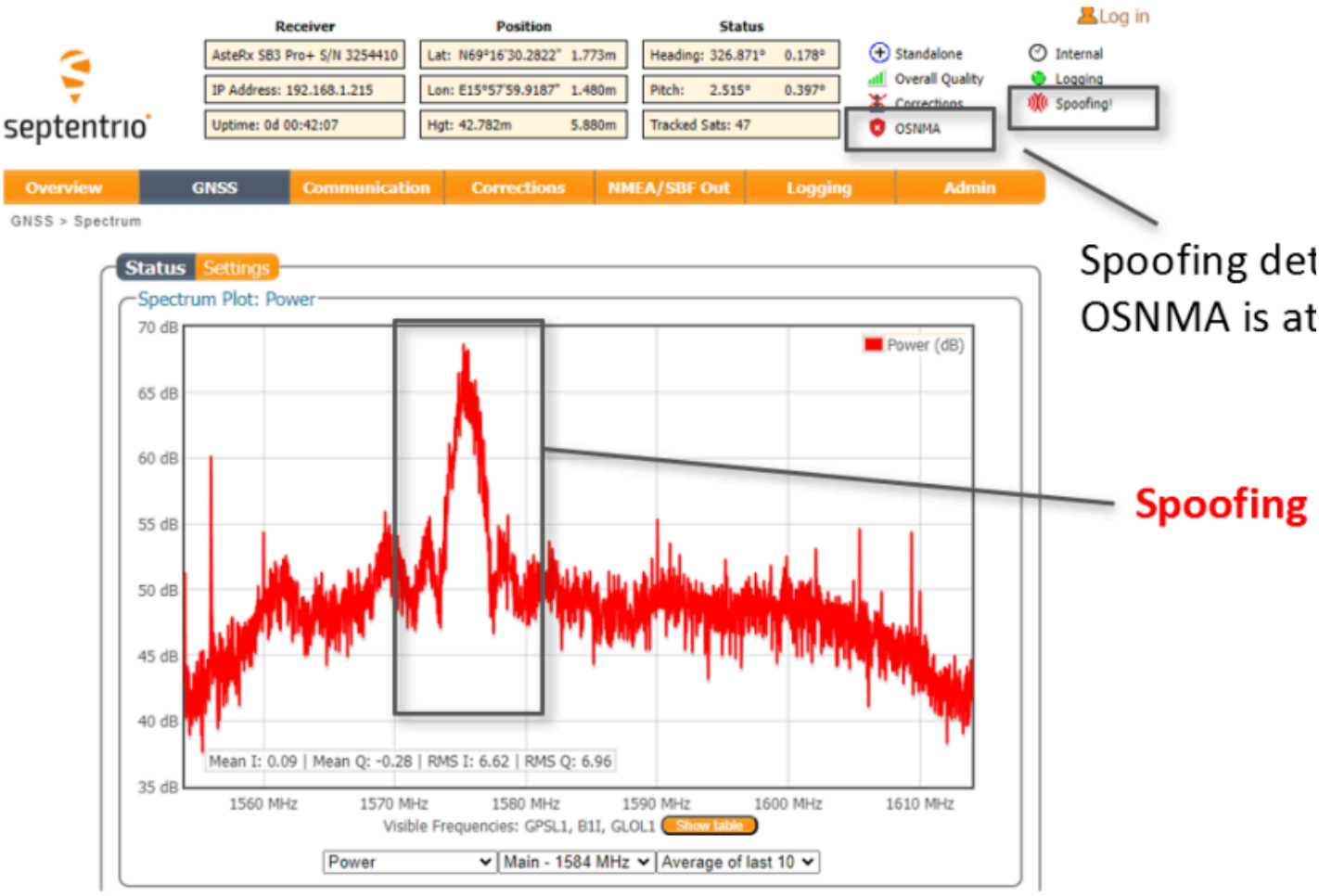

Figure 7: During this spoofing attack the OSNMA authentication mechanism from Galileo detects spoofing, as seen in the receiver's Web User I

While OS-NMA is effective for many types of spoofing it can still be fooled by "meaconing", which is a type of spo duplicates and retransmits genuine signals with a time delay. This is why NMA should be accompanied by other methods such as advanced anomaly detection for comprehensive spoofing security.

Dedication to continuous improvement and security

It may feel cumbersome to continuously update our computers and operating systems to maintain security, but recognize that regular updates are essential for ensuring a secure and resilient system. Similarly, Septentrio acti participates in events like JammerTest to validate its anti-jamming and anti-spoofing algorithms. These events al valuable data that enable us to continuously enhance our technology, achieving ever-higher levels of resilience.

** Jamming is a form of radio interference which occurs when GPS frequency is overpowered by other radio waves, resulting in accuracy degradatic loss of position. Spoofing is a malicious form of radio interference, where misleading signals are sent into the receiver, resulting in faulty coordinate target away from its predefined track.*

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